ARCHITECTURE REVIEW · CANDIDATE DESIGN EXERCISE

Reliability and asset-health data platform

Release 1 design, governance response, and the first twelve months. Presented to the architecture review board and the VP, Asset Management.

PRESENTED BY
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ROLE
Senior AWS Solutions Architect candidate

SESSION
15 min pitch, 15 min defense

Confidential. Candidate exercise, not a real client. All data synthetic.

PART 1 OF 9 · EXECUTIVE SUMMARY

One platform, three releases, trust built in order

For the VP, Asset Management. No service names on this slide.

Month 4

Release 1

Regulatory reliability report built from source data, not spreadsheets.

Month 7

Release 2

Ranked equipment risk list, ready before summer.

Months 6 to 12

Release 3

Crew notes structured. Live grid data piloted, only after Security signs off.

What it costs

Build, release 1	**$0.27M to $0.45M**
Build, all three releases	**$0.72M to $1.2M**
Run, steady state	**$16k to $57k a year**

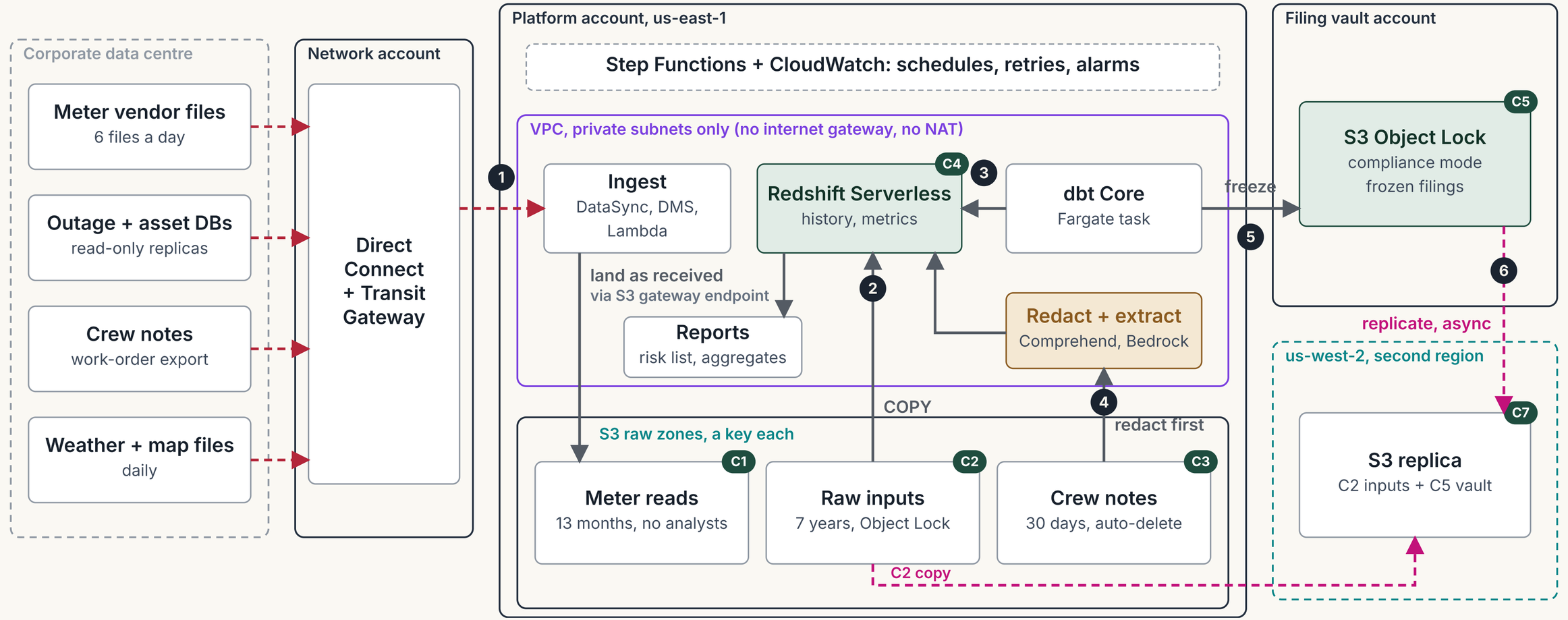
People at $12k to $20k per person-month, a planning rate. Cloud at list price.

Three decisions we need

1. Name an owner for last year's method by month 2. We cannot build the number without one.
2. Approve the target state and zone map, so release 1 starts in month 1.
3. Approve dbt Core, or name the fallback, by month 1.

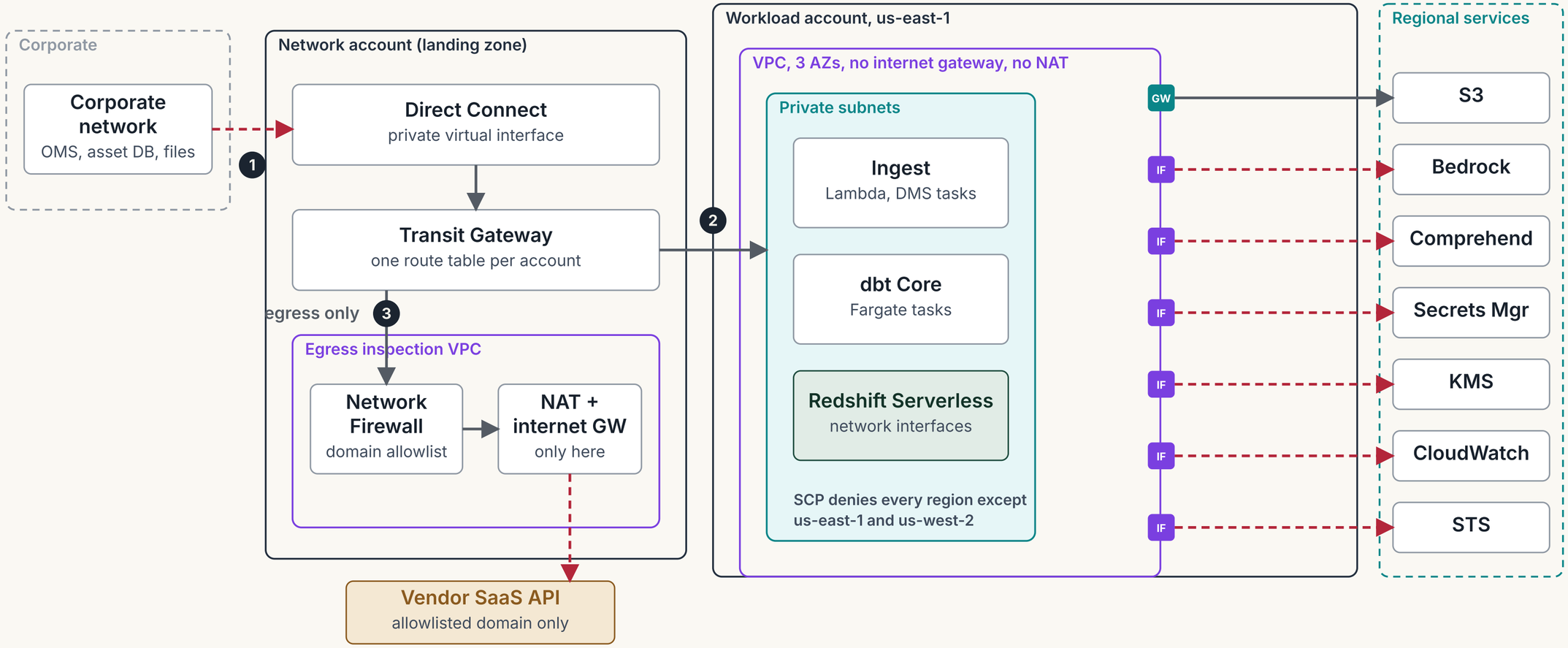
The chatbot question: not in release 1. A chat interface over data nobody can fully explain yet adds risk, not value. We revisit at month 9 at the earliest.

Two accounts, two regions, one direction of travel



C1 to C7 are the copies of customer data, listed on the next slide. The vault sits in its own account, owned by Compliance, so platform admins cannot edit filing evidence.

Network and identity: no internet path from the data



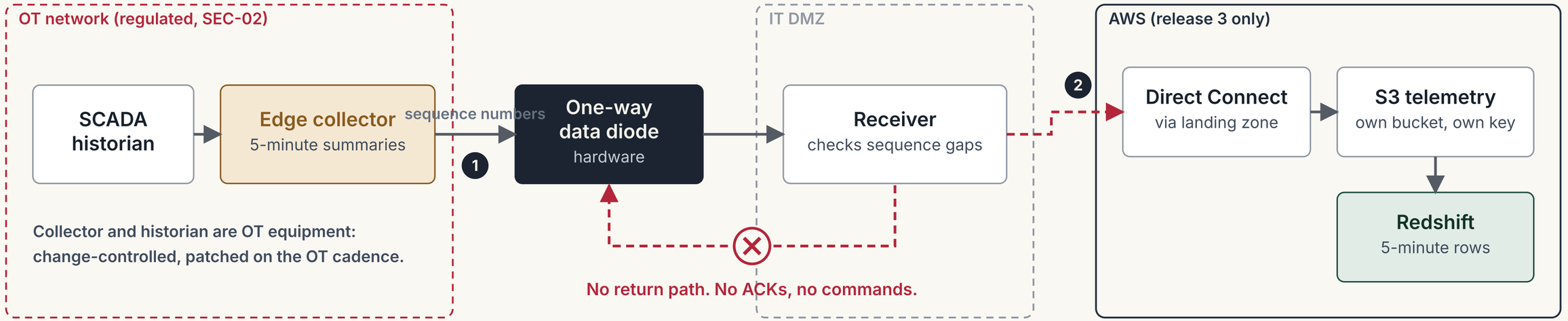
Squares on the VPC edge are endpoints: GW is S3 gateway, IF is interface. Bedrock is invoked in-region only. Sign-in via corporate IdP and Identity Center, no IAM users.

Every copy of customer data, named

ID	Copy	Holds	Kept	Controls
C1	Raw reads, S3	15-minute reads, pseudonymous meter ID	13 months	Own key, no analyst access, break-glass only
C2	Raw inputs, S3	Outages, assets, map, customer counts, weather. No names	7 years	Object Lock, pipeline roles only
C3	Raw notes, S3	Crew notes, may hold names and phones	30 days	Own key, auto-delete, redacted first
C4	Warehouse	Events, history, metrics, redacted notes, AI fields	7 yr regulatory, 3 yr other	Row-level rules, masking, aggregates by default
C5	Filing vault, S3	Frozen filing snapshots and evidence packs	7 years	Object Lock compliance mode, separate account
C6	Backups and logs	Snapshots, versions, access logs. No content	Logs 1 year	Same keys as source
C7	Second-region replicas	Copies of C2 and C5 in us-west-2	Same as source	Same controls. C1 is not replicated: the vendor can resend it

The medical or life-support flag is the most sensitive field in the brief. It never enters the platform, so it has no row here.

Grid data leaves OT one way, and only in release 3



What leaves

5-minute summaries only: average, min, max, switch count. About 14M rows a day from 50,000 tags.

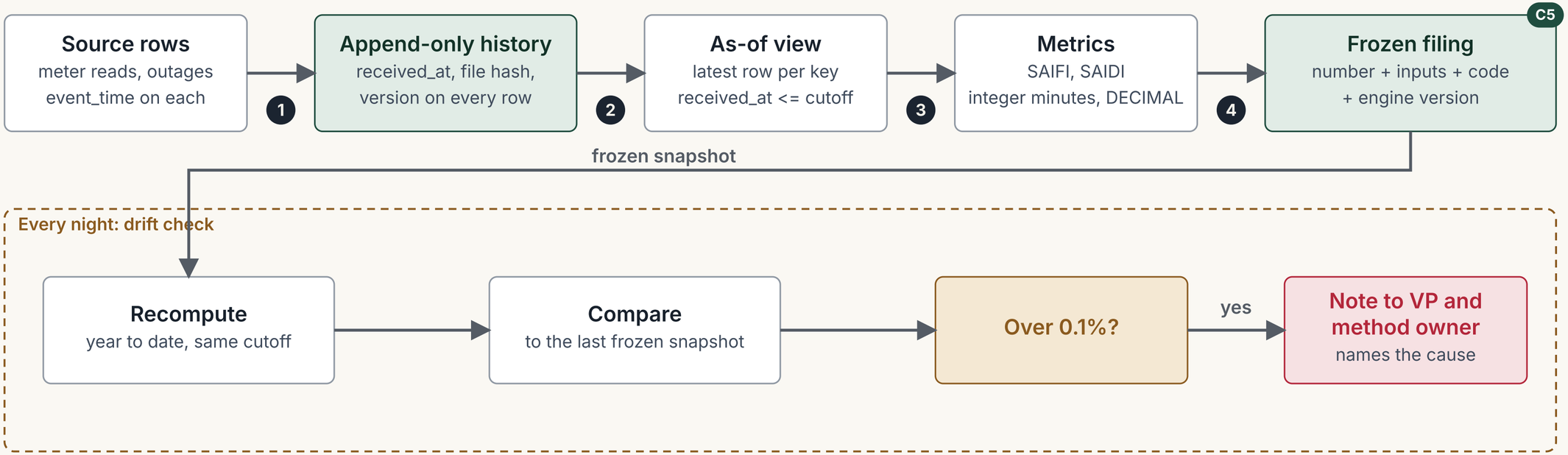
If data is lost

No ACKs means no resends. Sequence numbers expose gaps. Gaps are logged, never filled by guessing.

What this does not stop

A compromised collector could send bad data out. It is OT equipment, in scope for SEC-02, with OT-cadence patching.

Reproduce any filed number, two years later



Data flow 1 Step order

Late data

Reads arrive up to 72 hours late. A correction is a new row. Month end settles after 5 days.

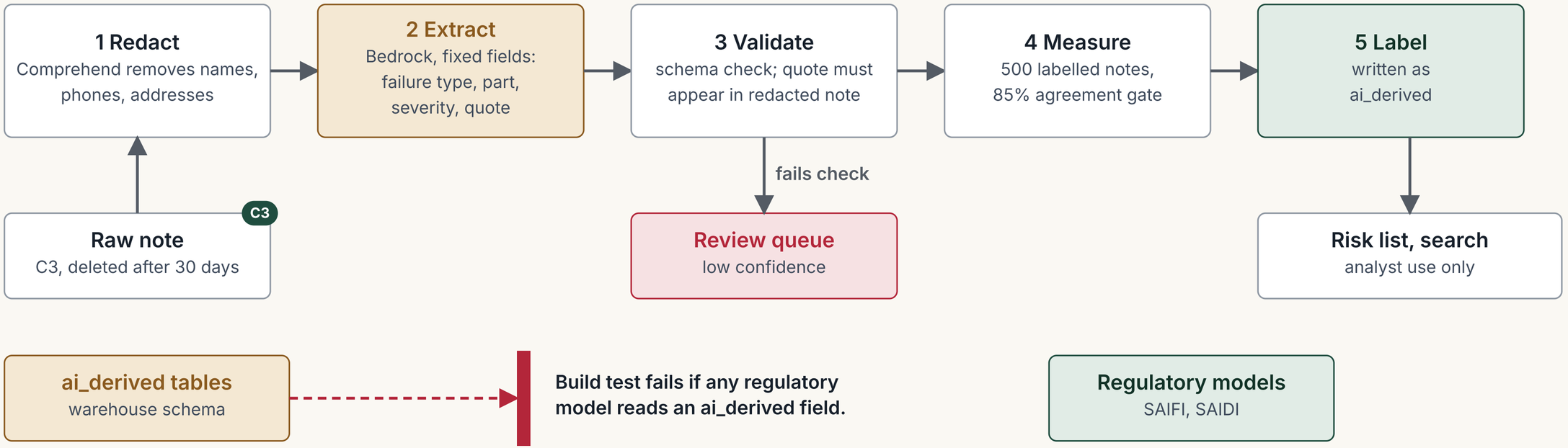
Wrong meter links (2 to 4%)

The outage record's customer count is the filed number. Meter events only check it. Disagreements go to a review list. The note shows the range with and without disputed links.

Same number, any day

Integer minutes and DECIMAL ratios, so a warehouse update cannot move a figure. Redshift Serverless cannot pin its engine, so the vault stores the output and we rerun last year's filing after every update.

AI only where it earns its place



Use	AI?	Why
Regulatory numbers, late data, link fixes	No	Must be exact and explainable. Plain SQL, fully tested.
Equipment risk, v1	No	A transparent score is enough. A learned model only if it beats the score on past failures.
Crew-note extraction	Yes	350,000 free-text notes a year. Text to failure type, part and severity.
Chatbot over notes	Not yet	Exposes personal data and gives answers nobody can check. Month 9 at the earliest.

One decision conforms to standards. One we hold firm on.

DECISION 1 · WAREHOUSE STANDARD

Conform to Redshift. No exception.

Context. Redshift is approved. Open table formats on S3 are under evaluation. Meter files grow about 440 GB a year.

Decision. Redshift Serverless is the only analytical store. S3 holds raw files as received, not managed tables.

Rejected. Open-table lakehouse (needs an exception and skills the team lacks). Provisioned Redshift (steadier at high use, we are far below it). Athena over S3 (splits the governance model).

Cost. Detailed meter reads are not queryable in release 1. Revisit at month 12.

DECISION 2 · HISTORY MODEL

Append-only history. Frozen filings.

Context. Reads arrive up to 72 hours late. A regulator may ask for a number two years from now.

Decision. Never overwrite. Every row keeps event time and received time. A filing run fixes a cutoff and writes to a locked vault.

Rejected. Update in place with snapshots (they expire). History tables only (show change, not what we knew when). Rebuild from raw files each time (slow, costly).

Cost. Only events and aggregates are kept in the warehouse, so storage grows under 5%. The real cost is a cutoff rule in every query, hidden inside views.

Board comments, answered by reference (1 of 2)

Ref	Our response	Residual risk we are not hiding	Owner
SEC-01	Cloud never connects into OT. Grid data leaves through a one-way hardware gateway, release 3 only. Physical pen test and a tabletop before go-live.	A fault loses data, it cannot open a way in. Tampering at the collector is OT-owned.	OT security lead
SEC-02	Release 1 touches no OT. In release 3 the historian, gateway and OT-side collector are all treated as regulated.	Compliance may rule summaries in scope. We ask first.	Compliance owner
SEC-03	Corporate sign-in only. Services use roles. No IAM users or long-lived keys.	One SaaS API credential is long-lived: read-only, 30-day rotation, alarmed outside the sync window.	Platform lead
SEC-04	Every copy listed with owner, retention and controls (C1 to C7). Macie scans the buckets.	Analysts could copy data off-platform. Exports are blocked and audited.	Data steward
LEG-01	Analysts see aggregates of 15 or more customers. Customer-level reads need a ticket and a time-boxed role.	15 is a starting value, not validated for rural feeders. Privacy officer confirms before go-live.	Privacy officer

Where we cannot fully satisfy a concern, we say so.

Board comments, answered by reference (2 of 2)

Ref	Our response	Residual risk we are not hiding	Owner
LEG-02	Notes redacted before any model. Bedrock stores nothing and does not train on inputs, per AWS terms. Terms re-checked at each model version.	Redaction can miss an unusual name. Sample audits, residual risk accepted by Legal.	Legal, data steward
LEG-03	All data stays in two US regions. Region-deny policy blocks the rest. Bedrock is invoked in-region, no cross-region profile.	None identified.	Cloud governance
EA-01	Conform. Redshift is the only warehouse. No exception requested (ADR-001).	dbt Core is a new tool. Approval requested, fallback is stored procedures.	Enterprise architect
EA-02	Use the landing zone for accounts, network, identity, logging and security tools. Two workload accounts only.	Shared-service changes follow their own process and add lead time.	Platform lead
EA-03	Every regulatory number has lineage: source hash, model, metric, in each filing pack.	Lineage covers our code. Upstream source edits are outside it.	Data steward

Four people, one on-call rotation, calm by design

2
things wake someone: a security alarm, or a failed filing run in the 10 days before a deadline

8 hrs
to load meter files, 99% of batches

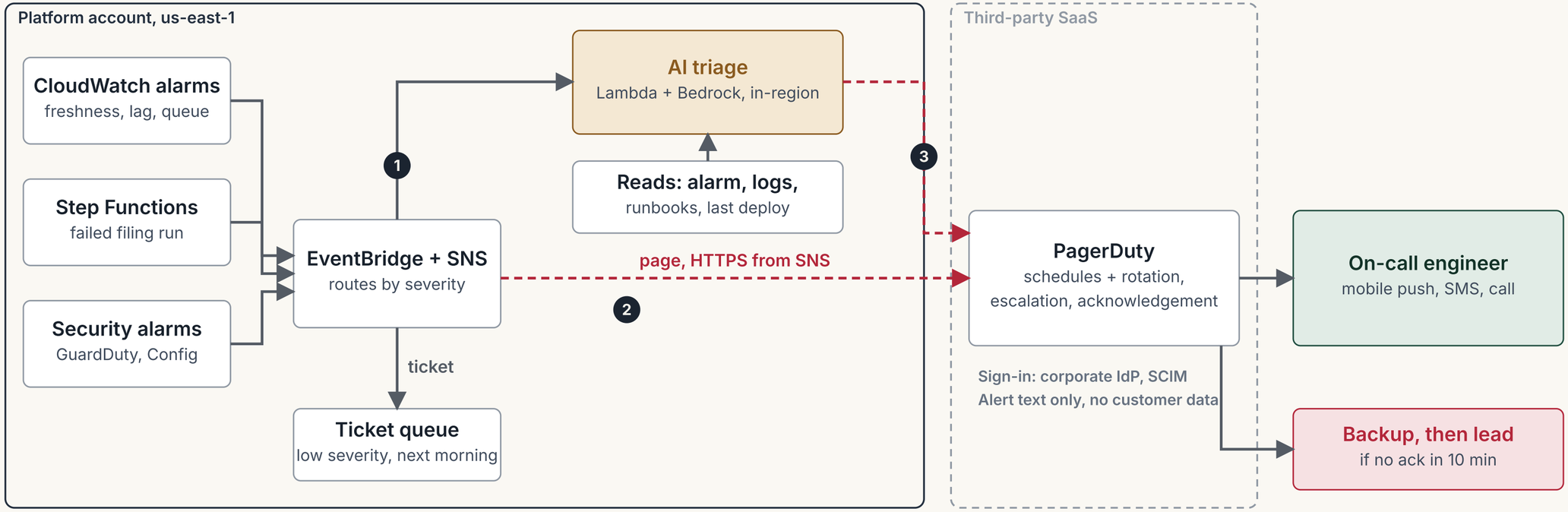
6 a.m.
ranked equipment list ready, every day

24 hrs
to recover from a lost region. Raw inputs and the vault are already copied.

What fails	How we know	Automatic response	Human action
Meter file late or missing	Freshness alarm at 8 hours	Retry the pull. Other sources continue	Morning ticket, ask the vendor
Source database unreachable	Sync-lag alarm	Retry with backoff for 6 hours	Check the link, then the DBA
A data test fails	Pipeline stops at that model	Quarantine the rows, alert	Steward fixes or overrides with a note
Model service throttled	Review-queue growth alarm	Back off and retry, queue is durable	None unless over 3 days
A whole region is lost	Platform health check	Redeploy from code in us-west-2, restore from replicas	Declare recovery, target 24 hours

Handover: two weeks shadowing, one runbook page per alarm, a game day before go-live. Infrastructure and alarms are code, reviewed like any change.

On-call: a paged human, a rotation, and an AI that does the reading



Who gets paged

Only the two wake-up conditions. Everything else is a ticket for the morning. Fewer pages, fewer ignored pages.

Rotation

Four people, weekly primary and a backup, kept in PagerDuty with phone push and escalation, not in a spreadsheet. Use the client's paging tool if one exists.

AI triage

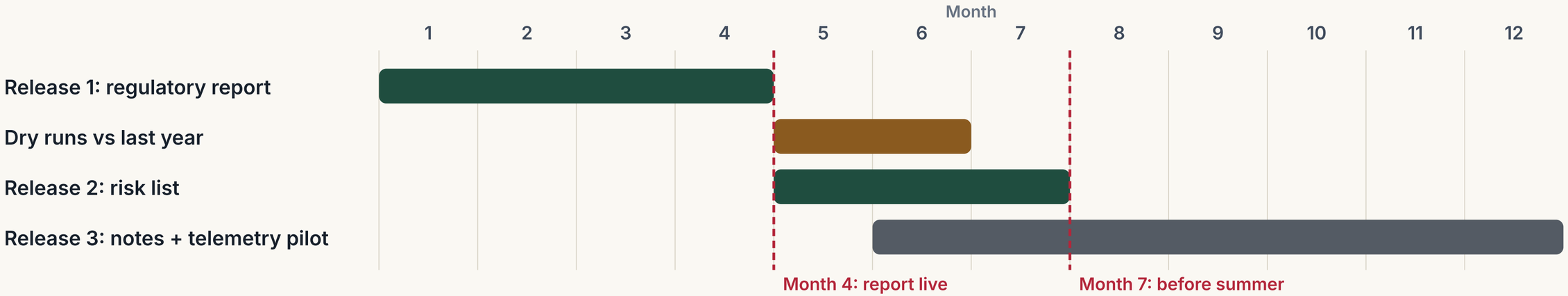
Reads the alarm, logs, runbook and last deploy, then adds a summary and a suggested step to the incident. Read-only: it suggests, a person acts.

What it costs, and in what order it arrives

$16k to $57k
cloud cost a year at steady state, about $1.4k to $4.7k a month

$0.27M to $0.45M
to build release 1: 22 person-months, a 5 to 6 person team over 4 to 5 months

$0.72M to $1.2M
to build all three releases: 60 person-months in 12 months



AWS list prices before discount. Warehouse capped at 32 compute units. Full working in the appendix. Swap in your own loaded engineer cost if you staff this internally: that moves the build range, not the run range.

CLOSE

What I need from the board today

1

Approve the target state and zone map

Three data zones by sensitivity, two US accounts, one warehouse. Release 1 starts in month 1.

2

Approve dbt Core, or name the fallback

The one new tool. Fallback is stored procedures with a generated manifest. Answer needed by month 1.

3

Agree the grid data path in principle

One-way gateway, 5-minute values. Nothing is built before release 3, and Security approves before any pilot.

Build trust first. Deliver the regulatory report. Expand only where the data and controls justify it.

APPENDIX

The working behind the numbers

Opened only if the board asks.

A · Assumptions and scope
B · Components and interfaces
C · Sizing and cost
D · Data model, tests, decisions, prototype, questions

APPENDIX A

Assumptions, and what release 1 leaves out

Assumptions the brief left open

- **Method.** Last year's ratios and storm-day rule are the reference. Parameterised, not redefined.
- **Customers served.** Year-end count unless last year used an average. One setting, signed off by the method owner.
- **Primary count.** The outage record's customers-affected number leads. Meter events and map links check it.
- **Retention.** Regulatory inputs 7 years, meter reads 13 months. Legal confirms.
- **Regions.** Two US regions. The second holds replicas of C2 and C5 only.
- **Prices.** AWS list prices. People rates are planning figures, not a quote.
- **Paging.** Client standard tool if one exists, else PagerDuty. Needs a Security review as a third-party service.

Left out of release 1 on purpose

- **Grid telemetry.** Path designed now, built in release 3.
- **Customer-level reads.** Kept in the raw store. Only events and aggregates enter the warehouse.
- **Personal details.** Names, addresses, contacts and medical flags never enter the platform.
- **AI.** Note extraction and the risk score arrive in releases 2 and 3.
- **Open table formats.** Not approved, so not used.

APPENDIX B

Components, responsibilities, interfaces

Component	Responsibility	Takes in	Hands on
Ingest DataSync, DMS, Lambda	Moves files and tables in, checked by checksum	Pushes from the corporate DC, read-only DB sessions	Raw store
Raw store S3, three buckets	Immutable landing, split by sensitivity	Ingest, over the S3 gateway endpoint	Warehouse load, AI step
Warehouse Redshift Serverless	Append-only history, metrics, marts, row-level rules	COPY from raw store, SQL from dbt	Reports, risk list, filing vault
Transform dbt Core on Fargate	SQL models, data tests, lineage graph	Warehouse tables	Run artefacts to the vault
AI Comprehend, Bedrock	Redact, extract, validate, queue for review	Raw notes bucket only	ai_derived schema, nothing else
Run Step Functions, CloudWatch	Schedules, retries, alarms, runbooks	Every component	SNS, then PagerDuty
On-call PagerDuty, AI triage	Pages, rotation, escalation; triage notes on the incident	Alarm text via SNS, logs, runbooks	Phone of the on-call engineer

Every interface is private, role-authenticated and logged. People: corporate IdP, SSO, one role per job. dbt fallback if not approved: stored procedures plus a generated manifest, same tests, same lineage output.

APPENDIX C

Sizing working

Item	Working	Result
Meter reads	2.1M meters × 96 reads a day	202M reads a day, 74B a year
Meter files	30 bytes a row, 5:1 compression, 6 files a day	1.2 GB a day, 440 GB a year
Corrections	1% of rows rewritten late	2M rows a day, appended never overwritten
Outage events	50M a year × 100 bytes	5 GB a year
Crew notes	350k work orders a year × 10 years × 2 KB	3.5M notes, 7 GB of text
Grid telemetry	50k tags, 5-minute summaries, 30 bytes a row	14M rows a day, 40 GB a year
AI per note	600 tokens in, 120 out, small model, plus redaction	About $0.002 a note. $0.7k a year. Backfill $7k to $20k
Warehouse	40 to 120 compute-unit hours a day at $0.375, cap 32	$450 to $1,350 a month

Volumes are planning assumptions, replaced with the client's real counts in week 1. Nothing here is near a managed-service limit, which is why one serverless warehouse is enough and Spark is not needed.

APPENDIX C

Monthly cloud cost, line by line

Line	Low	High
Warehouse	$500	$1,470
Object storage	$50	$100
Database sync	$60	$200
File transfer	$5	$10
Orchestration	$50	$150
Containers (dbt runs)	$20	$60
On-call tool + AI triage	$100	$350

Line	Low	High
AI extraction	$60	$250
Risk model	$10	$50
Network	$120	$300
Security tooling	$100	$400
BI tool (none if existing)	$0	$600
Test environment	$250	$700
Second-region copies	$30	$100

Total a month: **$1,355 to $4,740**

Steady state a year: **$16k to $57k**

Year 1 at 70% ramp: **$11k to $40k**

List prices before discount. Biggest swings: warehouse hours, whether a BI tool exists, test environment uptime. People cost is separate: 22.5 person-months for release 1, 60 for the year.

How an "as of" rerun works

```
select *
from meter_reads
where received_at <= :cutoff
qualify row_number() over (
  partition by meter_id, event_time
  order by received_at desc
) = 1;
```

event_time	when the meter measured it
received_at	when we got it
source_file_hash	which file it came from
version	which load wrote it
engine_version	recorded at run time

- Late data is appended, never overwritten. A rerun with the old cutoff ignores it.
- The cutoff, code version, engine version and output are frozen in the vault as proof.
- Redshift Serverless tracks the current release and cannot be pinned. So the figure never depends on the engine: integer minutes and DECIMAL ratios, plus a yearly rerun of the prior filing as a regression test.

APPENDIX D

Tests that stop a bad number

Test	Rule	If it fails
Freshness	Newest file under 8 hours old	Alarm, morning ticket
Row count	Within 0.5% of the vendor manifest	Stop, quarantine the file
Duplicate keys	None per meter and time, per load	Stop at that model
Meter links	Each meter linked once, disputes listed	Reported with the number
Reasonableness	Each metric within ±10% of last year	Written explanation required
Drift	Year to date within 0.1% of last frozen snapshot	Note to VP and method owner
AI wall	No regulatory model reads ai_derived	Build fails

APPENDIX D

Eight decision records, and what we rejected

ID	Decision	Rejected, and why
001	Conform to Redshift, no exception	Open table format on S3: not approved, not needed at this size
002	Bitemporal, append-only history	Overwrite in place: cannot reproduce a filed number
003	SQL-first with dbt Core (approval requested)	Spark: nobody on the team knows it, data is small
004	One-way hardware gateway for grid data	Firewall rules only: one rule error becomes a two-way path
005	Analysts see aggregates only	Masked customer-level data: masking mistakes still leak
006	AI limited to crew notes, outside the regulatory path	AI in the regulatory calculation: cannot be explained to a regulator
007	Two accounts, US regions only; vault in its own account	One shared account: platform admins could alter filing evidence
008	PagerDuty for paging and rotation, alert text only	SNS email alone: no rotation, escalation or acknowledgement. Self-hosted paging: one more thing for four people to run

APPENDIX D · PROTOTYPE

A rerun that matches, and a drift alert that fires

What it is

Python and DuckDB on synthetic data: 40 feeders, 10,000 meters, 600 outages. Six tests, same SQL pattern as the warehouse design.

```
python src/demo.py
```

Left out on purpose

Real ingestion, access control, the full test suite, any AI. It proves the one risky idea, not the platform.

Demo moment	Result
Filed number, cutoff 31 Jan	SAIFI 8.87, SAIDI 1,020 min
Late corrections arrive, rerun same cutoff	Identical
Rerun on latest data	SAIDI 1,036 (+1.6%), drift alert fires
Storm-day rule removed	SAIDI 1,529
Disputed meter links excluded	SAIDI 1,000 (about -2%)

All figures come from synthetic data, so they show behaviour, not real performance.

APPENDIX

Tough questions, short answers

Question	Answer
Why not an open table format?	Not approved, and 440 GB a year does not need it. Revisit at the month 12 standards review.
What if the gateway fails?	Grid data stops, nothing else does. Releases 1 and 2 never depend on it. The collector buffers locally and gaps are found by sequence number.
How do you know the AI is right?	500 labelled notes, an 85% gate, an evidence quote on every output, and a build test that keeps it out of every regulatory number.
Last year cannot be reproduced.	Correct, and we will not pretend otherwise. Dry runs in months 5 and 6 compare our output to last year's filed numbers and explain every difference in writing.
Budget is halved. What goes?	Grid telemetry first, then note extraction. Release 1 and the risk list stay.
Why a paid paging tool for four people?	Rotation, phone push and escalation are the whole job, and email cannot do them. If the client already has a standard tool, we use it. The SNS hook is the same.
Why a risk score, not ML?	A transparent score can be explained and tested by month 7. A learned model comes in v2, only if it beats the score on past failures.